AADYA KASHYAP

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EDUCATION

University of California, San Diego | B.S., Bioengineering (GPA: 3.654) **2026**

SKILLS

- **Programming & Modeling:** MATLAB, C++, Python, Simulink, Arduino, Signal Processing, Numerical Methods, Data Analysis Tools
- **Mechanical Design & Analysis:** SolidWorks, Fusion 360, AutoCAD, ANSYS
- **Prototyping & Manufacturing:** Resin 3D Printing, Laser Cutting, Basic Machining, Rapid Prototyping
- **Laboratory & Imaging:** ImageJ, Data Viewer, Biological Tissue Dissection (chicken & rabbit knees), Micro-CT, Protocol Development, Lab Environment Experience
- **Machining & Fabrication:** Drill Press , Drill Mill, Band Saw, Hand Tools
- **Technical Communication & Project Planning:** Technical Writing, Documentation, Gantt Chart, MS Office (Word, Excel, PowerPoint)

ENGINEERING EXPERIENCE

Biological Joint Biomechanics (Team 34 BJBM BENG 187B) **Sep 2025 - Present**

- Designed and prototyped a displacement-controlled mechanical compression device using SolidWorks & 3D printing in a lab environment to apply physiological loads to cartilage samples during micro-CT imaging.
- Converted anatomical alignment and imaging constraints into mechanical design requirements
- Developed repeatable loading protocols to reduce motion artifacts.
- Performed sample preparation, micro-CT imaging, and strain and proteoglycan analysis using ImageJ and Data Viewer, and documented protocols for future studies.

Biomedical Engineering Society (BMES), UCSD | Mechanical Subunit Member, Pump Team **Dec 2025 - Mar 2025**

- Designed and prototyped a pulsatile cardiac pump system in Fusion 360 using medical device design principles, simulating atrial and ventricular flow conditions.
- Developed a crankshaft-driven syringe mechanism with mitral and aortic valves to generate controlled, pulsatile flow.
- Generated DXF/STEP files for laser cutting and rapid prototyping of system components
- Fabricated and assembled tubing, reservoirs, and housing using bench saws, drill press, epoxy coating, and basic machining techniques.

Engineering World Health – UCSD | Prosthetic Arm Design **Sep 2024 - Dec 2025**

- Designed parametric CAD models of prosthetic finger assemblies in Fusion 360.
- Performed mechanical prototyping, fit and tolerance evaluation, and documented design changes to support iterative redesign cycles.
- Integrated an Arduino-controlled tendon-driven actuation system to evaluate finger joint kinematics and motion transmission.
- Developed and maintained a Bill of Materials (BOM) to manage actuators, tendons, fasteners, and printed joint components across multiple design iterations.

Electrocardiogram (ECG) Machine **Sep 2024 - Jun 2025**

- Built a device to record and display heart electrical signals using the AD8232 ECG sensor and TFT display.
- Designed and tested amplification and filtering circuits to improve ECG signal quality.
- Prototyped on breadboard and PCB applied HRV analysis to detect rhythm irregularities
- Analyzed experimental data using MATLAB to filter signals, quantify system performance metrics, and visualize results using data analysis tools.

Robot and Clock Design project | MAE 3 **Sep 2024**

- Created 3D CAD models in Fusion 360 and AutoCAD for robot prototyping, including gear mechanisms, rack-and-pinion assemblies, and lift system components
- Fabricated and refined components using drill press, bench saw, laser cutter, and 3D printer.
- Wrote technical reports documenting design intent, tolerancing, safety factor calculations, fabrication steps, and assembly testing.

AWARDS

- **Provost Honor:** Fall 2024, Winter 2025, Spring 2025, Fall 2025